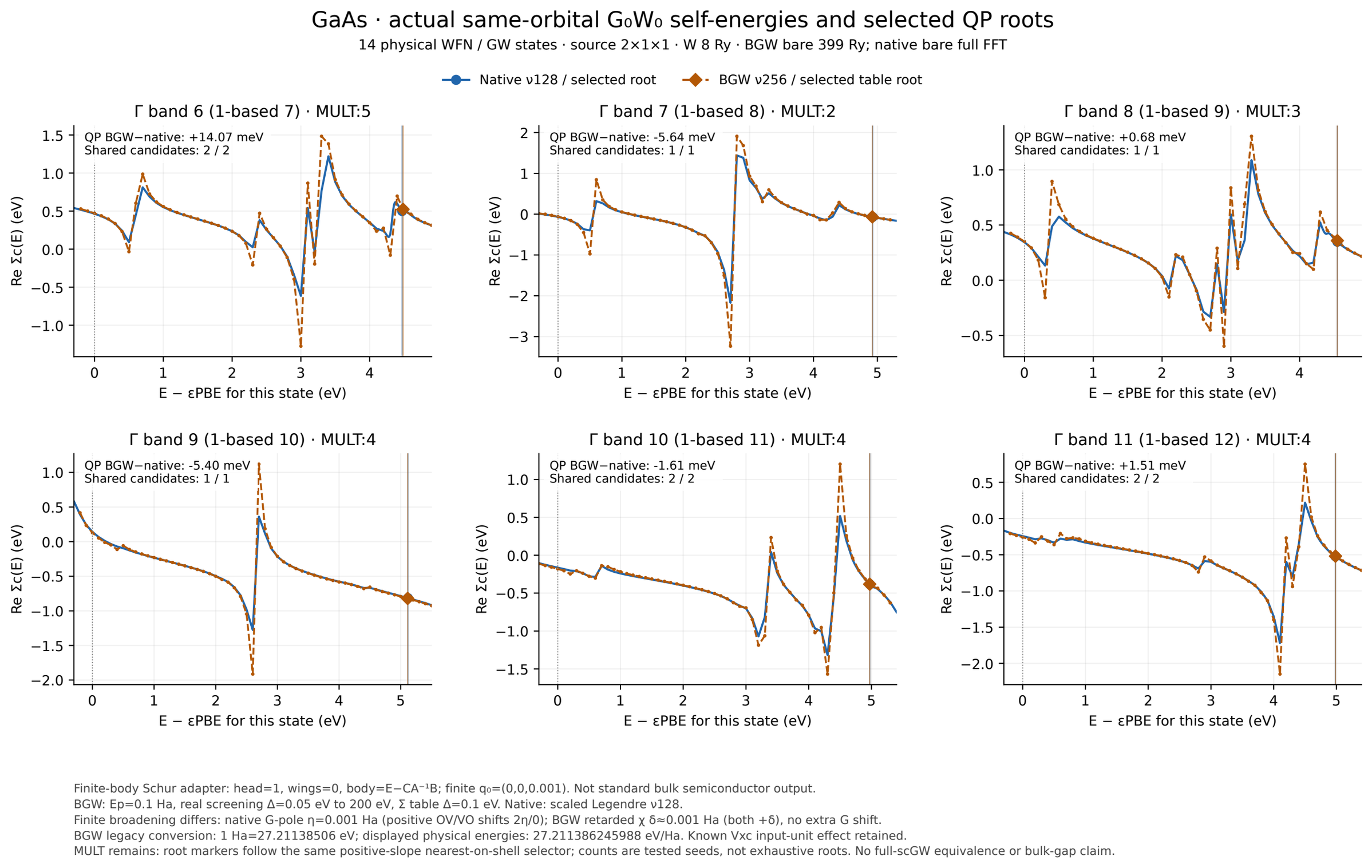
GaAs · actual same-orbital G_0W_0 self-energies and selected QP roots

14 physical WFN / GW states · source 2×1×1 · W 8 Ry · BGW bare 399 Ry; native bare full FFT



Finite-body Schur adapter: head=1, wings=0, body=E−CA⁻¹B; finite q_0=(0,0,0.001). Not standard bulk semiconductor output.
BGW: Ep=0.1 Ha, real screening Δ=0.05 eV to 200 eV, Σ table Δ=0.1 eV. Native: scaled Legendre ν128.
Finite broadening differs: native G-pole η=0.001 Ha (positive OV/VO shifts 2η/0); BGW retarded χ δ≈0.001 Ha (both +δ), no extra G shift.
BGW legacy conversion: 1 Ha=27.21138506 eV; displayed physical energies: 27.211386245988 eV/Ha. Known Vxc input-unit effect retained.
MULT remains: root markers follow the same positive-slope nearest-on-shell selector; counts are tested seeds, not exhaustive roots. No full-scGW equivalence or bulk-gap claim.